

# Session Objectives

- Students will be able to compare and contrast large, medium, and small arteries and veins.
- Students will be able to describe different capillary types and how their morphology varies according to their function.
- Students will be able to discuss the histology and function of lymphatic vessels.
- Students will be able to discuss the unique functions of endothelial cells.
- Students will be able to discuss microvascular components that regulate blood flow.

Source: Chapter 13, pp 400-408, 432-433. Cardiovascular System, Histology: A text and Atlas, Ross and Pawlina,



- GENERAL CONCEPTS: VESSELS

**Arteries and veins have three layers** (tunics)

1. **Tunica intima**: Innermost layer of endothelium, basal lamina of the endothelial cells, and subendothelial layer (loose connective tissue). Arteries: subendothelial layer of intima contains a sheet-like layer (lamella) of fenestrated elastic material (the internal elastic membrane) which lets substances diffuse through and reach cells deep in the vessel wall.
2. **Tunica media**: Middle layer, mainly circumferentially arranged layers of smooth muscle cells. Arteries: this layer is pretty thick and extends from the internal to the external elastic membrane. The latter is a layer of elastin that separates the tunica media from the tunica adventitia.
3. **Tunica adventitia** – outermost connective tissue layer; contains (among other things) vasa vasorum which supply blood to vessel walls



# ARTERIES

- Large artery (elastic artery): > 1 cm dia
- Medium artery (muscular artery): 2-10 mm
- Small artery: 0.1 – 2 mm
- Arteriole: 5-30  $\mu\text{m}$ , smooth muscle layer 1 -2 cells
- Capillary: 4-10  $\mu\text{m}$ , only endothelium



# Large Elastic Arteries

## Tunica Intima

- -**endothelium**: simple squamous
- -**subendothelium**: CT & elastin
- -**internal elastic lamina**: elastin: fenestrated sheets

## Tunica Media

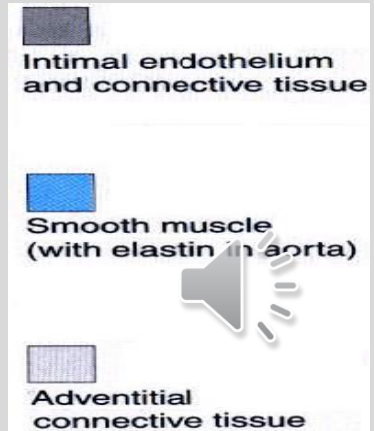
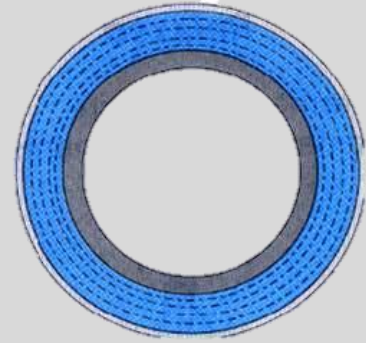
- -smooth muscle – spindle shaped cells
- -collagen fibers and ground substance made from smooth muscle cells, not fibroblasts

## Tunica Adventitia

- -**external elastic lamina**
- -collagen fibers, elastic fibers (not sheets)
- -fibroblasts and macrophages (main cells)
- -vasa vasorum and nervi vascularis

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# LARGE (ELASTIC) ARTERY

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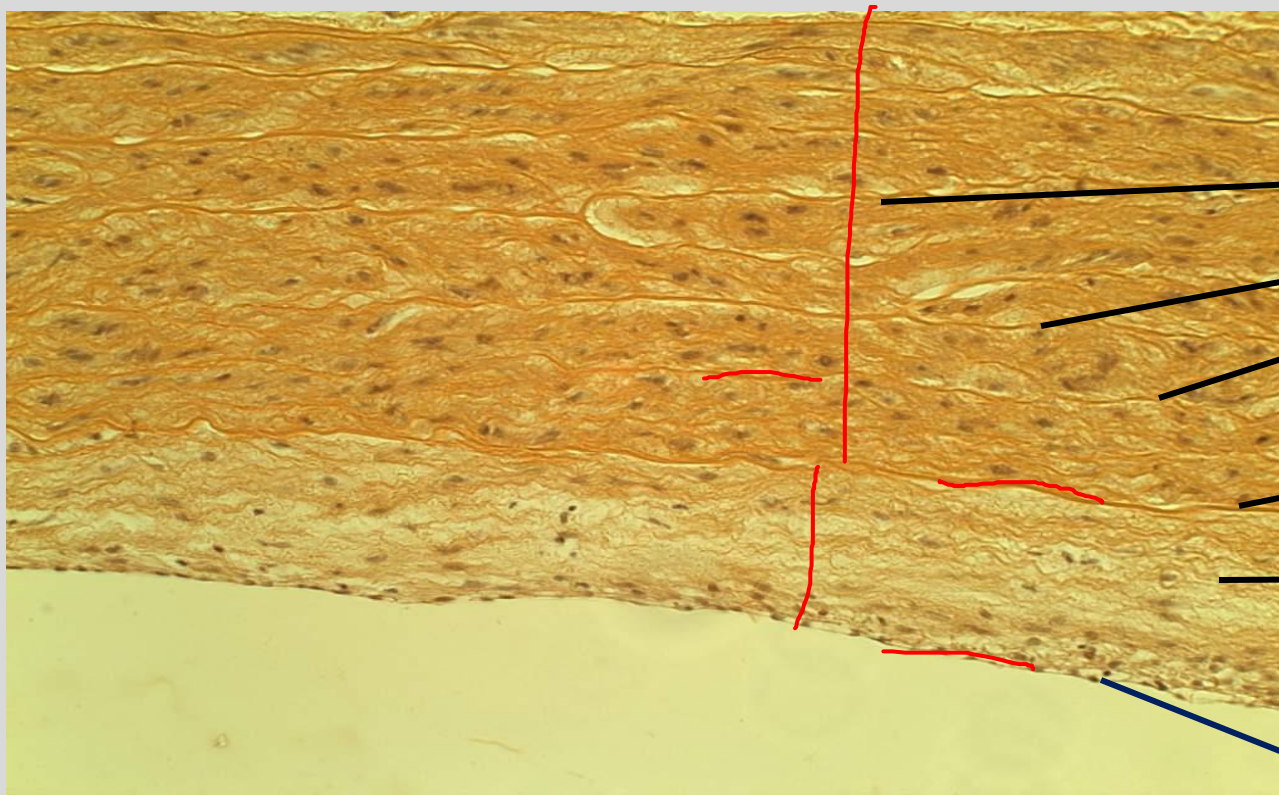
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*t. media*  
elastin sheets  
50-70 in aorta

*t. intima*  
internal  
elastic lamina

*t. intima*  
subendothelium  
connective tissue,  
elastin

Endothelium:  
simple squamous

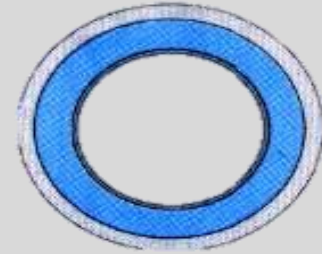


Source: Gerdes

# Muscular Arteries: most 'named' arteries of body

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tunica

- -endothelium

intima

- -subendothelium: sparse
- -internal elastic lamina: ~~PROMINENT~~

tunica

- -thickest layer

media

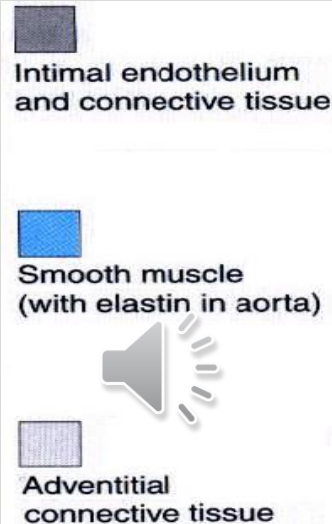
- -smooth muscle: **PREDOMINANT**
- -collagen with little elastin
- -external elastic lamina: in largest ones

tunica

- -collagen fibers, elastic fibers

adventitia

- -fibroblasts and adipose
- -vasa vasorum and nervi vascularis



# Muscular Artery

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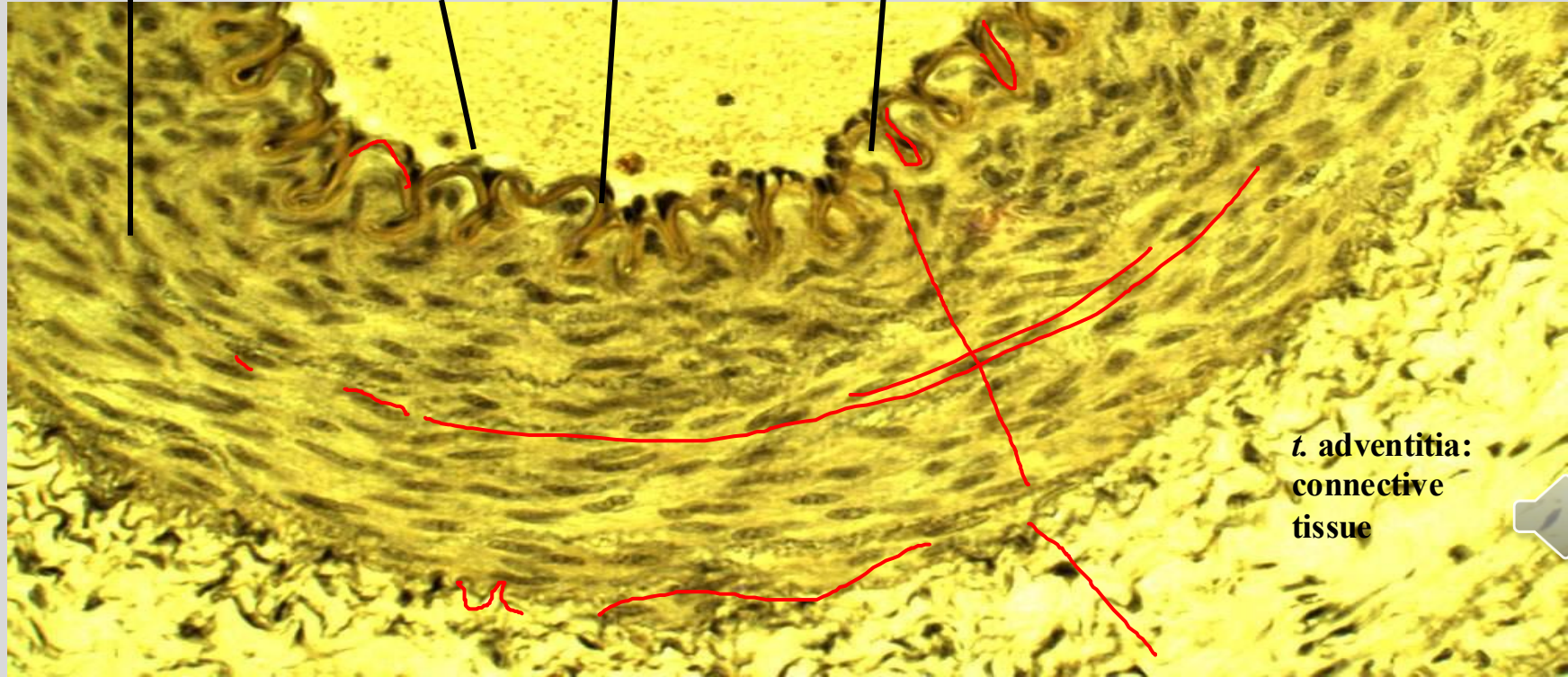
*t. media:*  
smooth muscle

*t. intima:*  
simple  
squamous  
endothelium

*t. intima:*  
internal elastic  
lamina

*t. intima:*  
subendothelium  
(sparse)

*t. adventitia:*  
connective  
tissue



# Small arteries to arterioles

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**tunica intima:**



Intimal endothelium  
and connective tissue

- endothelium
- internal elastic lamina: in larger ones

**tunica media:**



Smooth muscle  
(with elastin in aorta)

- smooth muscle:
  - ~8 layers in small artery
  - 1-2 smooth muscle layers in arteriole
- some collagen: in largest small arteries

**tunica**

**adventitia:**



Adventitial  
connective tissue

- connective tissue



arterioles regulate the flow of blood to capillaries

specialized arterioles can act as an A-V shunt



# Capillaries

**Tunica intima:** -one to three endothelial cells  
-basal lamina / basement membrane

*3 types--*

**continuous** pass metabolites and oxygen with tissues for waste and carbon dioxide (**muscle, lungs, CNS**); may have pericytes – see later discussion

**fenestrated** found in organs involved in absorption (intestine), filtration (kidney), and secretion (endocrine glands); have pinocytotic vesicles; pores/diaphragms

**discontinuous** a.k.a. **sinusoids**  
pass blood cells (liver, spleen, and bone marrow)



# Post-capillary venules and small veins

- **Tunica intima – endothelium**
  - Postcapillary venules in lymph nodes are called high endothelial venules (HEVs) because of prominent cuboidal endothelial cells
- **Tunica media**
  - **1-2 layers of smooth muscle** in small veins;
  - No tunica media present in postcapillary venules
  - Pericytes are present in post capillary venules
- **Tunica adventitia - connective tissue**
  - thin layer in small veins (not post capillary venules)
  - Respond to **histamine** (released by mast cells, ↑ capillary permeability) and **serotonin** (mostly vascular contraction)



# Medium veins

**Most 'named' veins in body; valves characteristic, especially in legs**

- **Tunica intima – endothelium**
  - **subendothelium** – thin, with occasional smooth muscle cells
- **Tunica media - smooth muscle (thinner than medium sized arteries)**
  - **collagen and fibroblasts**
- **Tunica adventitia - thickest layer in medium veins**
  - **elastic fibers longitudinal collagen & smooth muscle**



# Large veins

- **Tunica intima - endothelium**
  - thin subendothelium; boundary between tunica intima and media not often clear
- **Tunica media**
  - relatively thin with mostly circular smooth muscle, fibroblasts, and collagen
- **Tunica adventitia - thickest layer in large veins (e.g subclavian veins and venae cavae)**
  - longitudinal smooth muscle, fibroblasts, collagen, and elastic fibers

Note: Gravity can have effect on vein structure- Larger veins in head have thinner walls. Saphenous vein in leg has thicker muscle layer and is preferred for coronary bypass surgery because it is already more “muscularized”.



## Clinical alert about impaired microvascular blood flow:

**Ischemic heart disease:** obstructed coronary arteries

**Non-Ischemic heart disease:** Patent coronary arteries

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Many recent clinical reports show a high incidence of microvascular blood flow impairment in “Non-Ischemic” heart patients. This includes dilated heart failure (HFrEF), non-dilated heart failure (HFpEF) and diabetic cardiomyopathy. **So, many “non-ischemic” patients may actually have ischemia at the microvascular level.** See pdf review if interested in cardiology.

This has been under reported because the traditional method to measure this (PET scans) uses radioisotopes. So these measurements were rarely collected. Newer, non-radioactive technology is rapidly moving toward increased clinical use.

Animal research from my lab (AM Gerdes) suggests this may be due to reduced levels of cardiac tissue T3 (thyroid hormone), which regulates microvascular blood flow. See [pdf](#).



This will be an important area of clinical research to follow.

# Lymphatic Vessels

- Unidirectional, **convey fluid only FROM tissues**
- Smallest are **lymphatic capillaries**, numerous in loose connective tissue under skin epithelium and mucous membranes; **begin as “blind-ended” tubes** in microcapillary beds
- Lymphatic capillaries converge into larger lymphatic vessels which eventually unite to form two main channels, then empty into vascular system by draining into the large veins in the base of the neck
- **Lymph enters the blood at the junctions of the internal jugular and subclavian veins.** The largest lymphatic vessel, the **thoracic duct**, drains most of the body and empties into the veins on the left side. The other main channel is the **right lymphatic trunk**.



# Lymphatics, continued

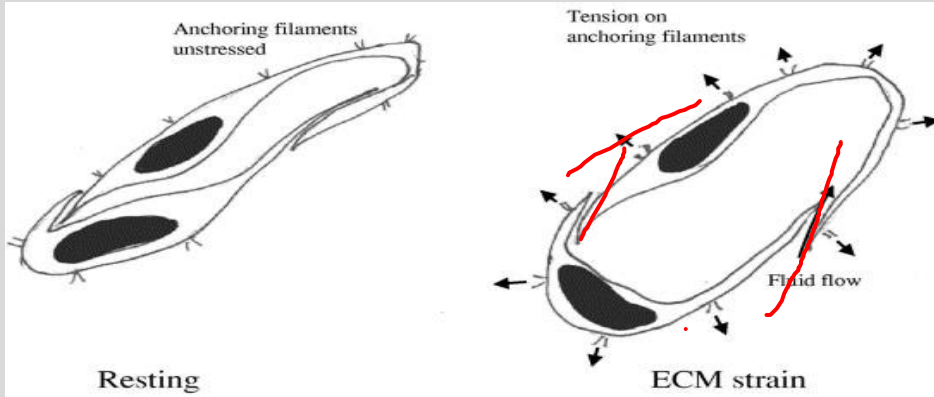
- **Greater permeability than blood capillaries**, more effective in removing protein-rich fluid from intercellular spaces
- Once fluid enters lymphatic vessel it is known as “**lymph**”
- Also serve to **transport proteins and lipids that are too large** to cross the fenestrations of the absorptive capillaries in the small intestine
- Before **lymph** returns to blood, it **passes through lymph nodes**, where it is exposed to cells of the immune system
- No central pump in lymphatic system. Valves help unidirectional flow and compression of skeletal muscles help move lymph along.



# Lymphatics, continued

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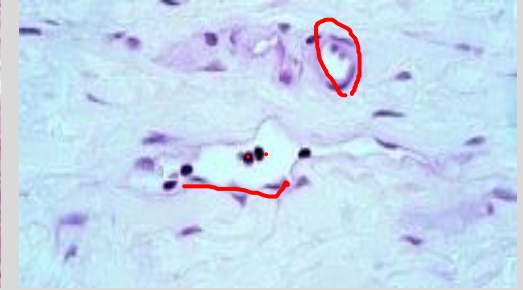
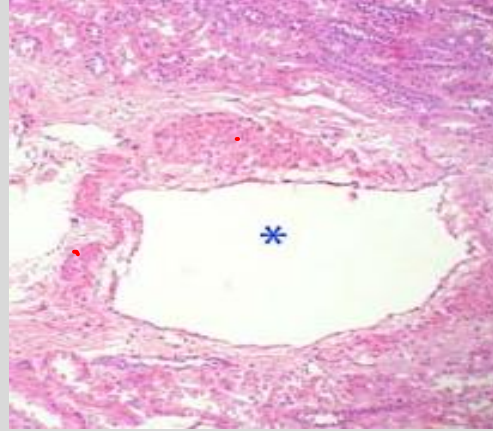
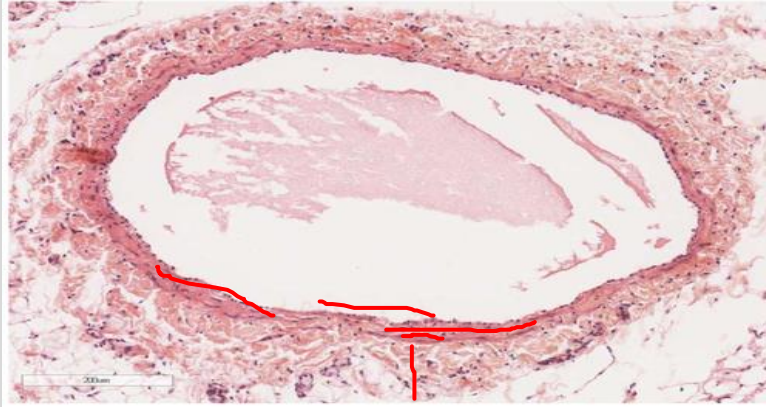
- **Lymphatic capillaries** = tubes of **endothelium which do not have continuous basal lamina** (which is why they are highly permeable) and more variable lumen diameters; **anchoring filaments** extend between the incomplete basal lamina and perivascular collagen to help keep the vessels patent when there is increased tissue pressure (e.g. inflammation)
- Lymphatic vessels progressively become thicker as walls develop more connective tissue and smooth muscle



# Lymph vessels

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## **ducts**

- Thoracic and right lymphatic ducts empty into neck veins. Larger lymphatic vessels have smooth muscle layer.

## **larger vessels**

- Like medium veins with less developed tunics
- bring lymph through lymph organs

## **capillaries**

- Endothelium with thin connective tissue layer
- dead-end near blood capillaries
- pick up extra-cellular fluid from tissues



## Summary Slide

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